

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Midterm Examination of Semester 2

Year: 2019

Subject: 341152 Electric Circuits II

Section: 5-6

Date: 22 January 2020

Time: 15.00-12.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. The test is divided into 3 sections. There will be 14 pages (including this page) and they are worth 65 points.

- Answer the questions on this test paper.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators can be used in this test.

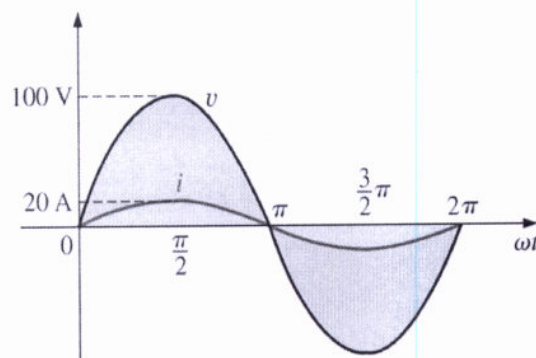
Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

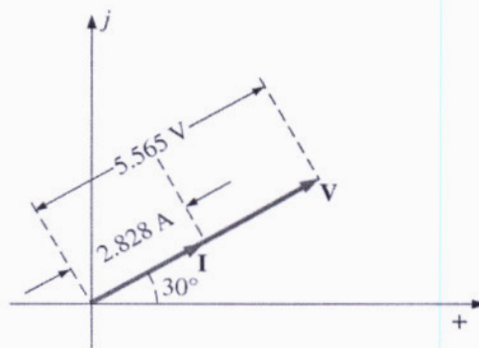
Part 1 (20 points) True/False

Instruction: Mark the correct answer for following questions in the provided answer sheet.

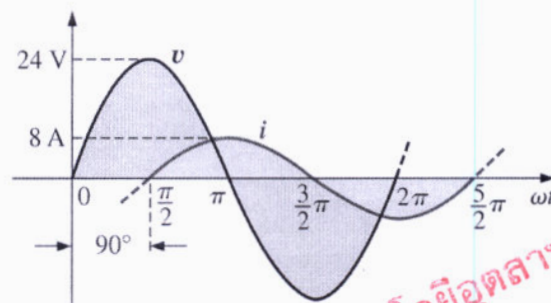
1. The direct current (DC) can be passed through the capacitor.
2. The insulating material in capacitor can be made from copper, aluminum, silver and etc.
3. In charging phase of capacitor, the current of capacitor decreases exponentially. This period is called transient state.
4. In transient state, time is less than the time constant (τ) of the system.
5. Steady state of the capacitor, the equivalent short-circuit represents the behaviors of capacitor.
6. In RL and RC networks, the energy will be fully stored after 5τ .
7. In RC network, the steady state is occurred at time of the switch is closed.
8. In charging state of capacitor, the voltage of capacitor increases exponentially as $v_c = V_s \times e^{t/\tau}$.
9. The time constant for a capacitor circuit equals its capacitance multiplied by its 2-times of the total resistance of the circuit.
10. The voltage and current characteristic of resistor is shown in figure below.



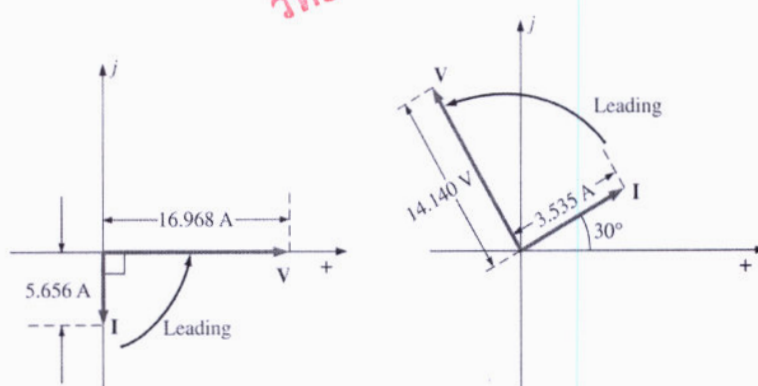
11. The voltage and current characteristic of resistor is shown in figure below.



12. The voltage and current characteristic of capacitor is shown in figure below.



13. The voltage and current characteristic of capacitor is shown in figure below.

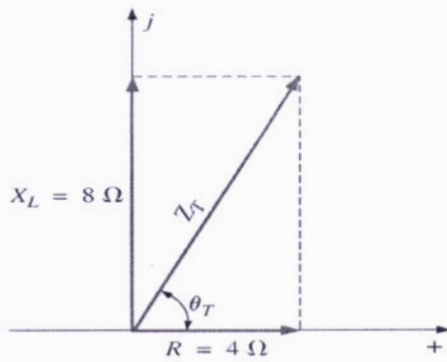


14. In practical, the inductor is usually made from copper. The copper coils therefore have internal resistance in Ohms.

15. The energy storage of inductor is proportional to the inductance value of the inductor as $W_L = \omega L$.

16. An ideal inductor looks like a short circuit to dc current.

17. The total impedance of phasor diagram in figure below is 12Ω .



18. The time constant for an inductor circuit equals its resistance divided by its inductance.

19. Power factor of resistive circuit is equal to zero.

20. Consider total power equation $P_T = EI \cos \theta_T$, the θ_T denotes the phase different between E and I .

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Part 2 (20 points) Multiple choices

Instruction: Mark the correct answer for following questions in the provided answer sheet.

1. What kind of capacitor is figure 1?

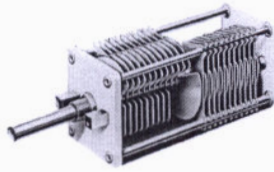


Figure 1

- (a) Ceramic trimmer capacitor (b) Electrolytic capacitor
(c) Mica capacitors (d) Air variable capacitor
2. Given the time constant and capacitance of $1\mu\text{s}$ and $1\mu\text{F}$, respectively. What is the resistance value of the RC circuit?
- (a) 0.1 Ohm (b) 1 Ohm (c) 10 Ohm (d) 100 Ohm
3. Find the capacitance of a parallel plate capacitor if the area of each plate is 0.075 m^2 and the distance between the plates is 1.77 mm. The dielectric is paraffin paper ($\epsilon_r = 2.5$) where $\epsilon_0 = 8.85 \times 10^{-12}\text{ F/m}$.
- (a) 0.522 pF (b) 522.45 pF (c) 0.937 pF (d) 938.05 pF
4. Find the capacitance of a parallel-plate air dielectric capacitor if the area of each of its plates is 0.1 m^2 and the distance between the plates is 2.0 mm.
- (a) 1.44 nF (b) 0.62 nF (c) 0.44 nF (d) 0.14 nF
5. What is the unit of inductance?
- (a) Ohm (b) Farad (c) Tesla (d) Henry

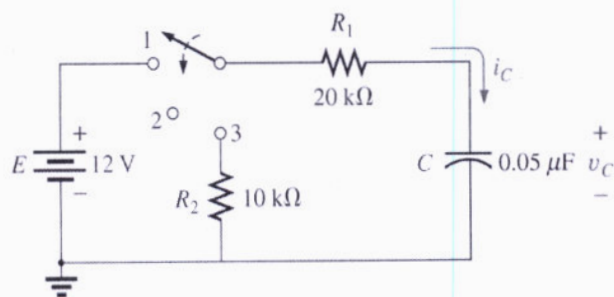


Figure 2

6. What is the time constant in the charging state of the circuit from figure 2?
- (a) 10 ms (b) 1 ms (c) 1 s (d) 1 μ s
7. What is the capacitor voltage at charging state in figure 2?
- (a) $12e^{t/\tau}$ [V] (b) $12e^{-t/\tau}$ [V] (c) $12(1 - e^{t/\tau})$ [V] (d) $12(1 - e^{-t/\tau})$ [V]
8. What is the capacitor current at charging state in figure 2?
- (a) $0.6e^{t/\tau}$ [mA] (b) $0.6e^{-t/\tau}$ [A] (c) $0.6e^{t/\tau}$ [A] (d) $0.6e^{-t/\tau}$ [mA]

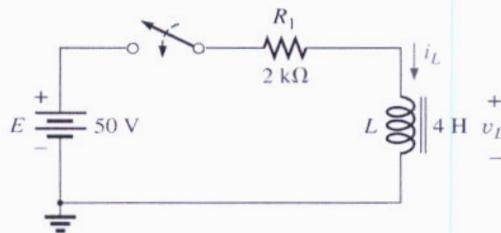


FIGURE 3

9. Find the time constant in of R-L circuit from figure 3.
- (a) 0.2ms (b) 2ms (c) 20ms (d) 200 ms
10. Find the voltage of inductor of R-L circuit at 1τ in figure 3.
- (a) 18.4V (b) 1.84V (c) 20V (d) 50V
11. Find the voltage of inductor of R-L circuit at 4τ in figure 3.
- (a) 0.95 V (b) 9.5 V (c) 10.5 V (d) 11.5 V
12. What is the inductive reactance at 400 Hz of a 4 mH inductor?
- (a) 12Ω (b) 20Ω (c) 25Ω (d) 0.2Ω

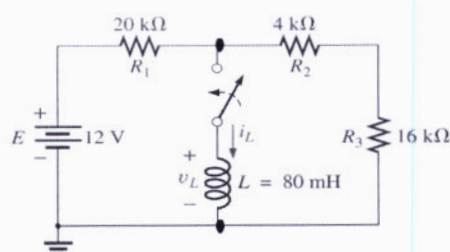


FIGURE 4

13. what is the time constant of the circuit from figure 4?
- (a) 8 μ s (b) 0.8 μ s (c) 11 μ s (d) 1.1 μ s

14. In steady state, what is the inductor current of the circuit from figure 4?
 (a) 0.6 mA (b) 6 mA (c) 4 mA (d) 0.4 mA
15. In steady state, what is the inductor voltage of the circuit from figure 4?
 (a) 12 (b) 1.2V (c) 0.5V (d) 0V
16. What is the time domain equation of inductor voltage (v_L) and current (i_L) from figure 5?

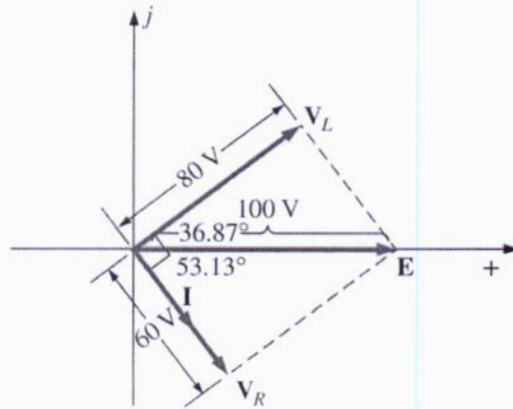


FIGURE 5

- (a) $v_L = \sqrt{2} \times 80 \sin(\omega t + 36.87^\circ)$ (b) $v_L = 80 \sin(\omega t + 36.87^\circ)$
 (c) $v_L = \sqrt{2} \times 80 \sin(\omega t - 36.87^\circ)$ (d) $v_L = 80 \sin(\omega t - 36.87^\circ)$
17. What is the time domain equation of resistor voltage (v_R) from figure 5?
 (a) $v_R = 60 \sin(\omega t - 53.13^\circ)$ (b) $v_R = \sqrt{2} \times 60 \sin(\omega t - 53.13^\circ)$
 (c) $v_R = 60 \sin(\omega t + 53.13^\circ)$ (d) $v_R = \sqrt{2} \times 60 \sin(\omega t + 53.13^\circ)$
18. What is the phasor form of voltage (v) in figure 6?

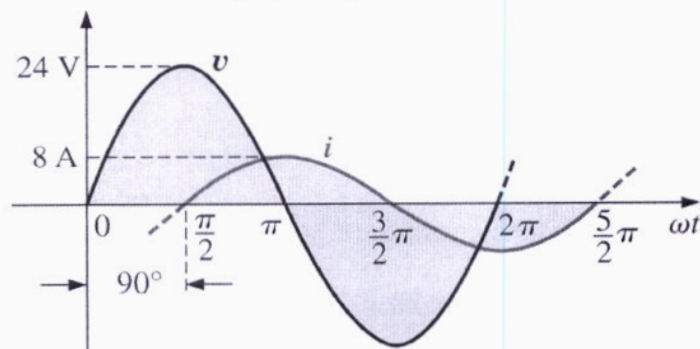


FIGURE 6

- (a) $V = \frac{24}{\sqrt{2}} \angle 0^\circ$ (b) $V = \sqrt{2} \times 24 \angle 90^\circ$ (c) $V = 24 \angle 0^\circ$ (d) $V = 24 \angle 90^\circ$

19. What is the phasor form of current (I) in figure 6?

- (a) $I = \frac{8}{\sqrt{2}} \angle 90^\circ$ (b) $I = \frac{8}{\sqrt{2}} \angle -90^\circ$ (c) $I = \sqrt{2} \times 8 \angle -90^\circ$ (d) $I = \sqrt{2} \times 8 \angle 90^\circ$

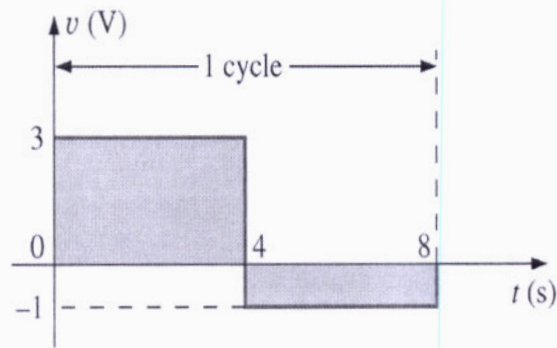


FIGURE 8

20. Find average value of signal in figure 8.

- (a) 2.0 V (b) 1.0 V (c) 1.5 V (d) 0.5 V

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Part 3 (25 points) Calculation

Instruction: Show the mathematical expression and answer of following problems.

1. Use figure 9 to find inductor voltage V_L and current I_L at 1τ to 6τ of the charging state and time constant.

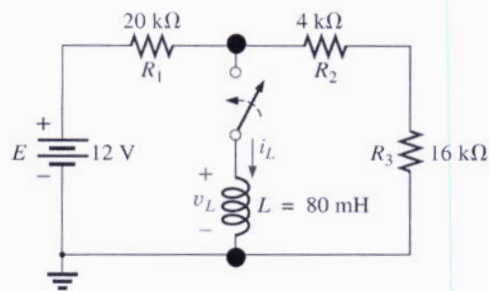


Figure 9

$\tau =$

V_L and I_L at $1\tau =$

V_L and I_L at $2\tau =$

V_L and I_L at $3\tau =$

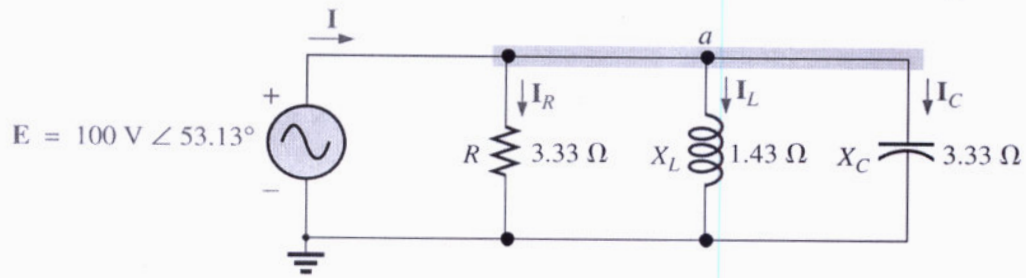
V_L and I_L at $4\tau =$

V_L and I_L at $5\tau =$

V_L and I_L at $6\tau =$

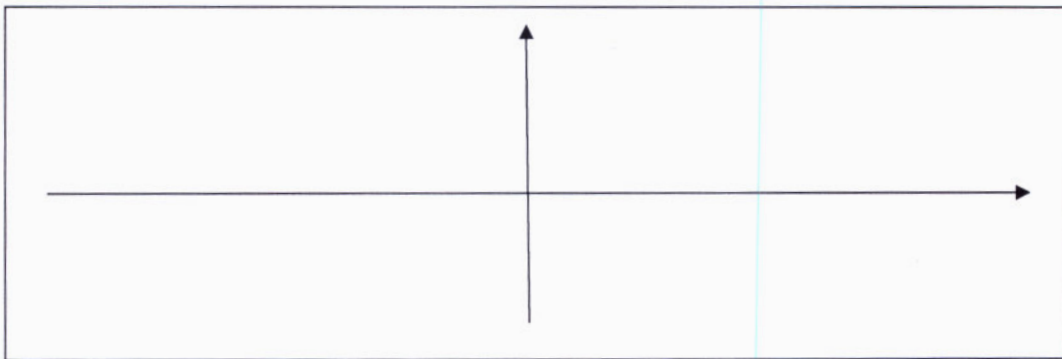
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2. Calculate the currents I_R , I_L and I_C of Figure 10 and draw the phasor diagram.



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Plot the Phasor diagram.



3. From circuit of Figure 11.

a) Find total IMPEDANCE (Z_T)

b) Find total current I_s .

c) Plot the Phasor diagram of voltage source, total current.

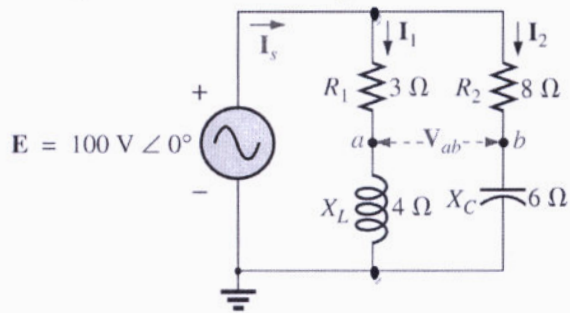
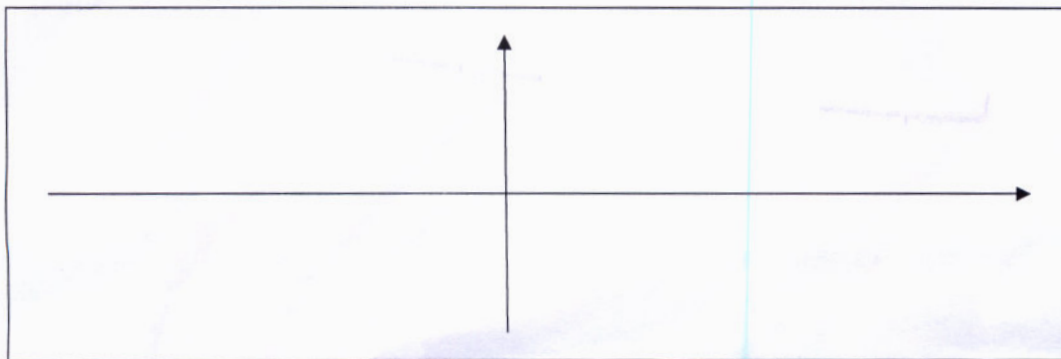


Figure 11

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Plot the Phasor diagram.



4. In the circuit for Figure 12, find V_L and V_C . Given current source $I = 10\sin(\omega t - 30^\circ)$ V. Plot the Phasor diagram of total current and voltage.

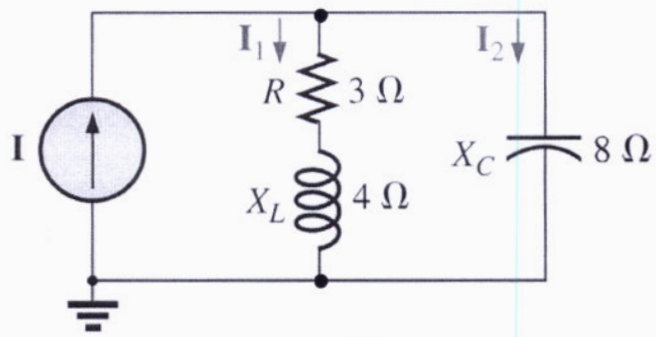
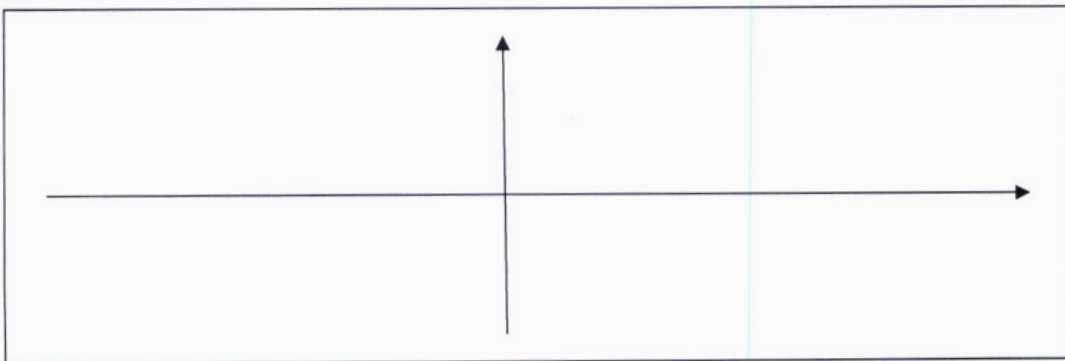


Figure 12

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Plot the Phasor diagram.



5. For the network of Fig. 13. Given frequency = 10 MHz
- Compute V_L .
 - Plot the Phasor diagram of V_L and I .

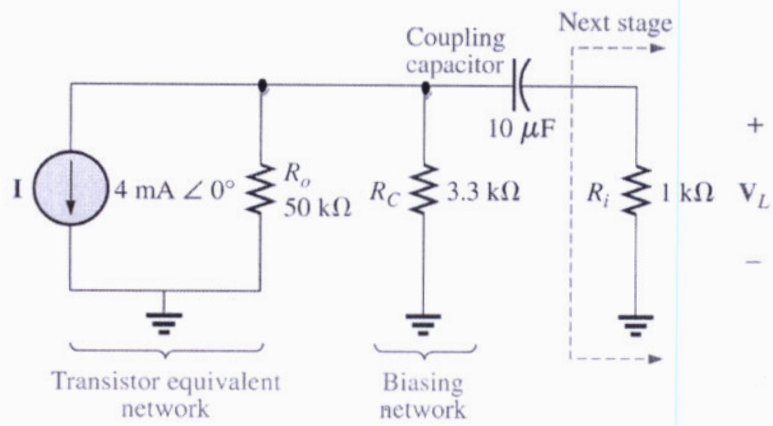
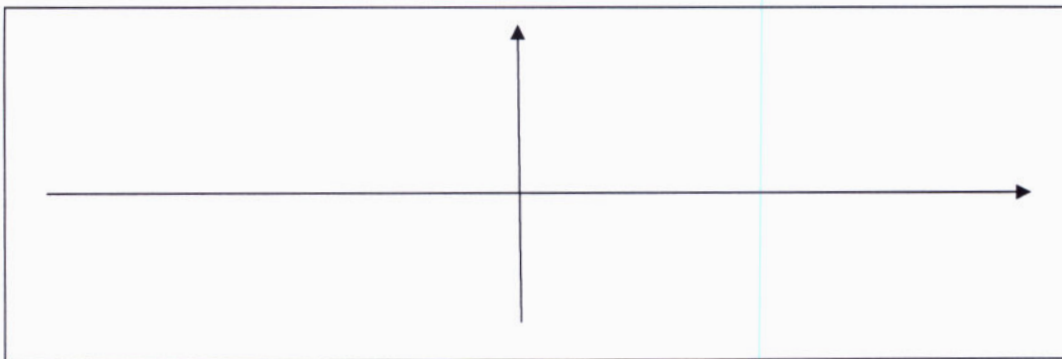


Figure 13

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Plot the Phasor diagram.



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Asst.Prof.Dr. Kittisak Phaebua

Answer Sheet

Name: _____ ID: _____

Subject: 341152 Electric Circuits II

Part 1

	True	False
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		
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18.		
19.		
20.		

Part 2

	a	b	c	d
1.				
2.				
3.				
4.				
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19.				
20.				

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